

SIDDHANTA MOHANTY

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EDUCATION

The Pennsylvania State University
Bachelor of Science in Computer Science

University Park, PA
Aug. 2022 – May 2026

- **Dean's List:** Fall 2023, Spring 2024, Fall 2025, Spring 2026 **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Computer Organization, Theory of Computation, Deep Learning for Computer Vision

EXPERIENCE

GenAI Intern

HCL Technologies

May 2025 – Aug. 2025
Redmond, WA

- Built a **RAG-based support automation system** using Flask, LangChain, and FAISS that analyzed production error logs, retrieved similar incidents and internal documentation, generated remediation recommendations, and sent automated Slack alerts, reducing support resolution time by **50%**
- Designed **retrieval pipelines** using FAISS and structured metadata to index internal documentation and historical incidents, improving the relevance of support recommendations
- Orchestrated **multi-agent workflows** across APIs, file systems, databases, and Confluence using **MCP** and LangChain for automated knowledge retrieval and actions

Research Assistant

Penn State University

Sep. 2025 – Dec. 2025
University Park, PA

- Designed **Reinforcement Learning** approaches to improve two-LLM code verification pipelines through iterative prompt refinement and compiler feedback
- Designed integration strategies for combining **RL** with validation tools (**Klee**, **CodeQL**) to enable automated code evaluation, self-correction and self-learning

Data Analytics Intern

Elevatoz Loyalty

June 2024 – Aug. 2024
Bengaluru, India

- Applied **SQL** and **Python (Pandas, NumPy)** to clean, transform, and analyze structured datasets.

PROJECTS

Self-Heal System | *Flask, LangChain, FAISS, PostgreSQL, OpenAI, Slack*

[Source Code](#)

- Built an **AI-powered self-healing backend** that captured database exceptions, embedded runtime error logs into FAISS, retrieved relevant historical incidents using semantic search with similarity thresholds, and generated context-aware explanations and remediation recommendations.
- Developed a **Slack-integrated incident response system** that delivered severity-based alerts, LLM-generated diagnostics and fixes, and a chatbot with access to recent error logs, database statistics, and a read-only SQL interface restricted to **SELECT** queries.

Prompt Tracer | *JavaScript, Chrome Extension API, Manifest V3*

[Source Code](#)

- Developed a **prompt observability Chrome extension** that analyzes and scores prompts in real time using a deterministic 5-point rubric, injecting a live feedback panel into ChatGPT, Claude, Grok, and Gemini for instant, consistent scoring and suggestions
- Built a dual-mode prompt optimization engine combining a privacy-safe rule-based optimizer with an optional OpenAI-powered rewrite pipeline, using ambiguity detection to distinguish prompts requiring clarification from those needing structural improvements

yourAlbrief | *Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, Resend, Vercel*

[Demo](#) | [Source Code](#)

- Built a full-stack **AI newsletter platform** with magic-link authentication, personalized topic preferences and delivery schedules, aggregating content from 50+ trusted sources and tracking per-issue email open and click analytics.

Swipeflix | *React, Vite, Tailwind CSS, TMDB API, Vercel*

[Demo](#) | [Source Code](#)

- Built a swipe-based **movie/TV discovery app** that personalized recommendations using a weighted scoring algorithm (genre match, popularity, recency, and format) informed by a 5-question onboarding quiz and live TMDB API data, while engineering custom gesture-driven swipe interactions with hand-built CSS transforms (no animation libraries) and session recaps highlighting swipe statistics and inferred viewing preferences.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks/Libraries: Flask, FastAPI(REST APIs), Django, Next.js, Node.js, React, LangChain, SQLAlchemy

AI/ML Tools: RAG, LLMs, Model Context Protocol (MCP), NumPy, PyTorch

Databases: PostgreSQL, MySQL, MongoDB, Vector Databases (FAISS, Pinecone)

Developer Tools: Git, GitHub, Docker, Prisma, Postman, AWS, Google Cloud Platform, VS Code, ROS2